# 1 description

Let $p$ denote fraction of infected patches and $N$ denote population density of infected patches.

$$\begin{tikzcd} \boxed{P_I} \arrow[r, shift left, "\text{infection}"] & \boxed{P_S} \arrow[l, shift left, "\text{fizzle or burnout}"] \end{tikzcd}$$

Figure 1: patch state transition

**spread:** Hosts move around and simultaneously bring pathogens to susceptible patches (combining movement and infection in one step to avoid calculating average density again)

$$f_s : \begin{pmatrix} p \\ N \end{pmatrix} \to \begin{pmatrix} p + (1 - \mathrm{e}^{-Bp})(1-p) \\ N + c(1-p)(S-N) \end{pmatrix}$$

$B$ is infection rate (currently setting $B$ as constant since we've already set $R_0$ as constant too), $c$ is moving probability of individual rats, and $S$ is the population density of susceptible patches (assumed constant).

**epidemic:** Infected hosts die and so some infected patches burnout.

$$f_e : \begin{pmatrix} p \\ N \end{pmatrix} \to \begin{pmatrix} P_1 p \\ (1 - zD)N \end{pmatrix}$$

$P_1$= `P1_prob(N)` is the persistence probability, $z$ is final size of this epidemic wave, $D$ is the death probability of infected hosts.

**growth:** Assume a constant birth rate $r$.

$$f_g : \begin{pmatrix} p \\ N \end{pmatrix} \to \begin{pmatrix} p \\ (1+r)N \end{pmatrix}$$

The overall iteration process is

$$f = f_g \circ f_e \circ f_s : \begin{pmatrix} p \\ N \end{pmatrix} \to \begin{pmatrix} P_1(p + (1 - \mathrm{e}^{-Bp})(1-p)) \\ (1+r)(1-zD)(N + c(1-p)(S-N)) \end{pmatrix}$$

where $P_1$= `P1_prob(N+c(1-p)(S-N))`

# 2 range of parameters

We want to make sure that $f([0,1]) \subset [0,1]$. The function attains its maximum at $x^* = \frac{k+1}{2k}$. Note that $f(0) = 0$ and $f(1) = 1$, so we'll need to require that $x^* \geq 1$, i.e. $0 \leq k \leq 1$.

- **infection**: $B > 0$

- **burnout**: $p = P_1 p$, $0 \leq P_1 \leq 1$
- **epidemic**: $0 \leq z \leq 1$ and $0 \leq D \leq 1$
- **movement:** $0 \leq c \leq 1$
- **birth**: $(1+r)(1-zD) \leq 1$

# 3 Disease Free Equilibrium

If we start at a point where $p > 0, N > 0$, all variables remain positive during the iterations as each step maps positive values to a positive values.
If we start at $p = 0, N > 0$ then population size ends up at

$$N_t = N_{t+1} = (1+r)(1-zD)(N_t + (p-1)(S-N_t))$$

i.e.

$$N_{DFE} = \frac{c(1+r)(1-zD)S}{1-(1+r)(1-zD)(1-c)}$$

$N_{DFE}$ increases with $r, c, S$ and decreases with $z, D$.

# 4 invasion analysis

$$J_{f_s} = \begin{pmatrix} (1+(1-p)B)\mathrm{e}^{-Bp} & 0 \\ -c(S-N) & 1 \end{pmatrix}$$

$$J_{f_e} = \begin{pmatrix} P_1 & p\frac{\partial P_1}{\partial N} \\ 0 & 1-zD \end{pmatrix}$$

$$J_{f_g} = \begin{pmatrix} 1 & 0 \\ 0 & 1+r \end{pmatrix}$$

And at DFE we have $p = 0, N = N_{DFE}$ so the matrices above are all lower triangular, so eigenvalues of $J_f = J_{f_g} J_{f_e} J_{f_s}$ are $\lambda_1 = (1+B)P_1$ and $\lambda_2 = (1-zD)(1+r)$. We required previously that $\lambda_2$ is always smaller than 1. If $(1+B)P_1|_{N=N_{DFE}+c(S-N_{DFE})} = (1+B)P_1|_{N=\frac{cS}{1-(1+r)(1-zD)(1-c)}} < 1$, DFE is stable, so the disease cannot persist; if it is greater than 1, DFE is unstable. Simulation shows that the disease will persist and system ends up at endemic equilibrium.

# 5 Endemic equilibrium

In the hazard model version the endemic equilibrium is given by

$$\begin{aligned} &P_1(p + (1-\mathrm{e}^{-Bp})(1-p)) = p \ (p \neq 0) \\ &(1+r)(1-zD)(N + c(1-p)(S-N)) = N \end{aligned}$$

where

$$P_1 = p_k(1 - q(x_{in})^{y\star(N+c(1-p)(S-N)})$$

$p_k, q(x_{in}), y\star$ are all defined in burnout paper. This is equivalent to

$$1 - \frac{N(t-1)}{c(S-N)} = p_k(1 - e^{y_\star t \ln q(x_{in})N})(1 - \frac{N(t-1)}{c(S-N)}e^{-B(1-\frac{N(t-1)}{c(S-N)})})$$

We can't have an analytic solution even with $W_r$ function (although we can always solve it numerically). (haven't proved EE is stable; we can always compute the eigenvalue numerically. I think there is a more elegant approach to prove it's stable but I just don't know ...

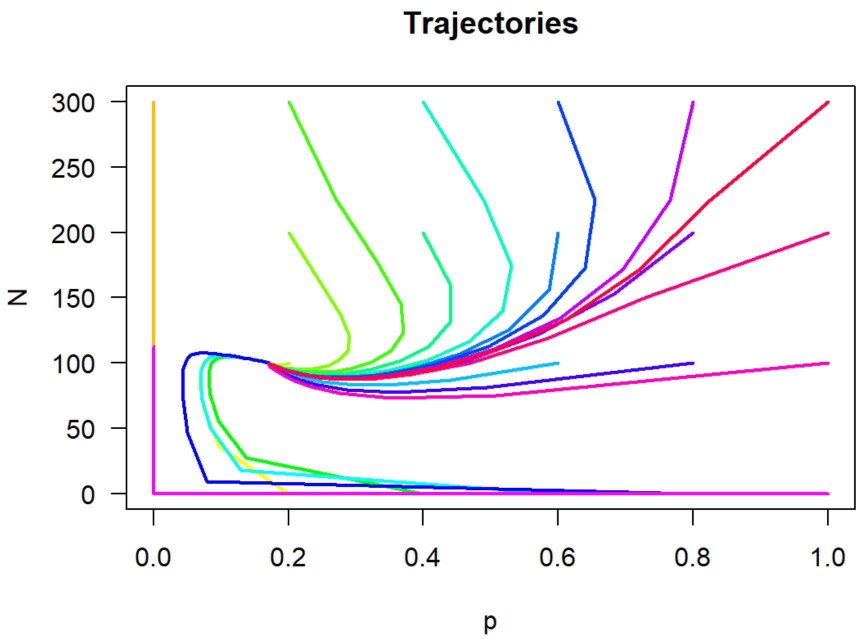


Figure 2: unstable DFE

# 6 Order of events

We have $f = f_g \circ f_e \circ f_s$ and this leads to

$$N_{DFE} = \frac{c(1+r)(1-zD)S}{1-(1+r)(1-zD)(1-c)}$$

and persistence condition

$$(1+B)P_1|_{N=\frac{cS}{1-(1+r)(1-zD)(1-c)}} > 1$$

We can do the same thing for $f_1 = f_s \circ f_g \circ f_e$ (cyclic permutation) and get

$$N_{DFE_1} = \frac{cS}{1-(1+r)(1-zD)(1-c)}$$

and threshold for persistence is the same

$$(1+B)P_1|_{N=N_{DFE_1}} = (1+B)P_1|_{N=\frac{cS}{1-(1+r)(1-zD)(1-c)}} > 1$$

Equations for endemic equilibrium of $f_1$ is a little bit more complicated than $f$ but we could also expect equivalent result since $\lim_{n\to\infty} f^n = \lim_{n\to\infty} f_1^n \circ f_s$

However if we consider non-cyclic permutation, for example, $f_2 = f_g \circ f_s \circ f_e$, then

$$N_{DFE_2} = \frac{cS(1+r)}{1-(1+r)(1-zD)(1-c)}$$

And persistence condition is

$$(1+B)P_1|_{N=N_{DFE_2}} > 1$$

$N_{DFE_2} > N_{DFE_1}$ and $P_1$ increases with $N$, so the disease becomes easier to persist. This makes sense biologically: with this order infected hosts breed before they die, and larger population size makes pathogens more likely to survive. (The previous order seems more reasonable than this one, as rats die very quickly after being infected.)

# 7 effect of R0

According to the threshold, larger $B$ and larger $P_1|_{N=\frac{cS}{1-(1+r)(1-zD)(1-c)}}$ is favored, and $P_1$ increases with $N$, i.e., increases with $r, c, S$ and decreases with $z, D$. However, $P_1$ increases with $R_0$ so seems that smaller $R_0$ would not be favored.

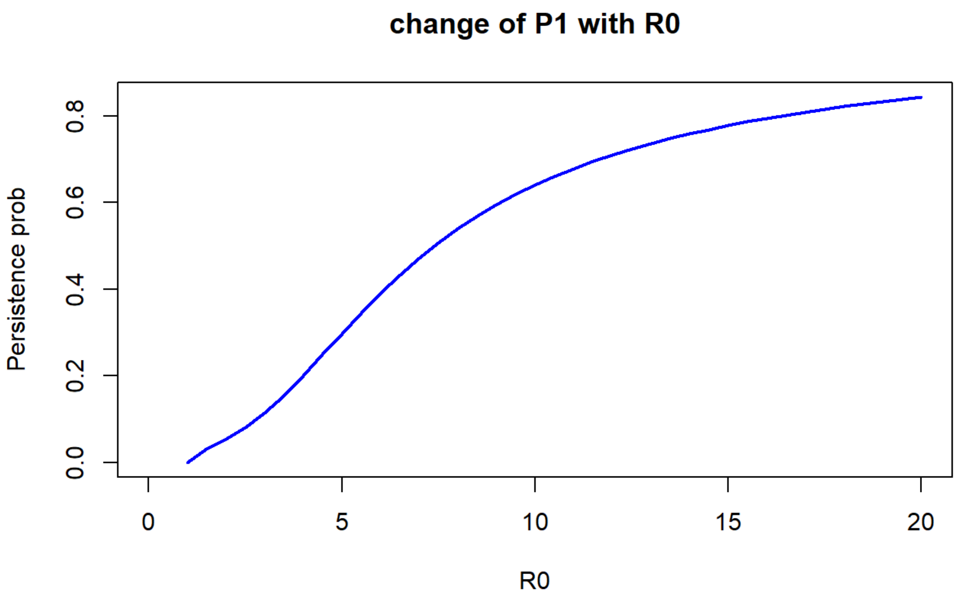


Figure 3: P1 increases with R0

It might be the decrease of $D$ that was favored in the evolution.

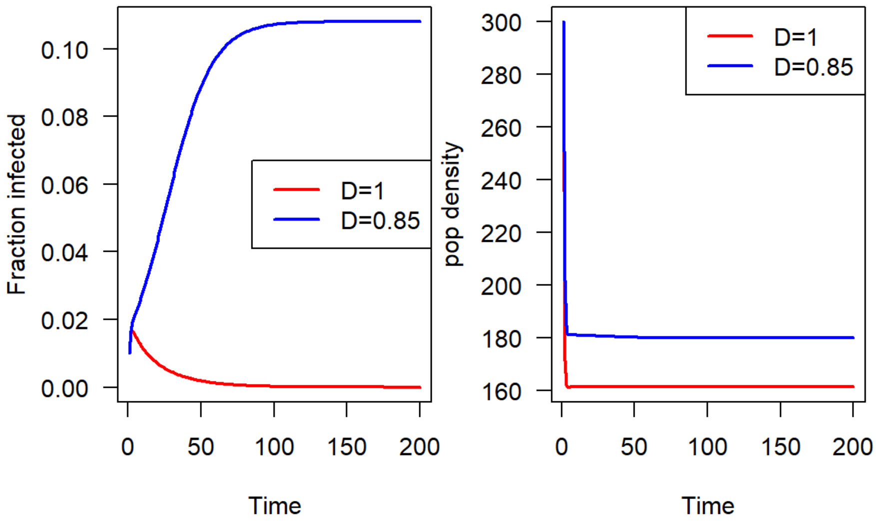


Figure 4: decrease in D makes disease persist